



KLSIC

Liquid-Phase Silicon Carbide

KEY PROPERTIES

- High hardness
- High temperature stability
- High thermal shock resistance
- High corrosion resistance
- High mechanical strength

APPLICATIONS

- Wear parts
- Semiconductor industry
- Ballistic protection

MAIN PROPERTIES

Property	Value	Unit
Density	3.24	g/cm ³
Flexural strength	530	MPa
Hardness	20	GPa
Fracture Toughness	5.8	MPa·√m
Young modulus	395	GPa
Thermal Conductivity	80	W/m·K
CTE (20–1000°C)	4.0	x10 ⁻⁶ K ⁻¹

* All properties measured at 20°C unless otherwise stated.

** Measured in a wall thickness of 2 mm

OPERATING CONDITIONS

Max working temp (air)		1800 °C
Recomended temp		<1600 °C
Continuous temp		1500 °C

ELECTRICAL BEHAVIOR

Semiconductor

MATERIAL INFORMATION

Material type		Silicon Carbide (liquid-phase sintered)
Structure		Dense ceramic
Key feature		High thermal + mechanical performance
Machinability		Low
Color		Dark grey

Local overheating may occur at electrical contact points due to current concentration and geometry effects.

The information provided is for guidance only and does not constitute a guarantee.

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